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# Blind reconstruction of BCH codes based on candidate generator polynomial in random situation

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A novel method for blind reconstruction of binary Bose–Chaudhuri–Hocquenghem codes is proposed. Compared to previously reported works, a new approach to find the goal generator polynomial is employed. First, using the feature that each codeword polynomial of a  $t$ -error-correcting Bose–Chaudhuri–Hocquenghem code has the same  $2t$  consecutive roots over Galois field, a new set of candidate generator polynomials is introduced. Then, this set in a random situation to find the correct generator polynomial is investigated. Monte Carlo simulations demonstrate the superiority of the proposed reconstruction algorithm compared to the previous methods.

**Introduction:** In non-cooperative communication situations such as eavesdropping and monitoring, obtaining channel encoder parameters is a key point [1–3]. During the last decades, many research studies have focused on the problem of blind reconstruction, namely, binary cyclic codes in which the aim is to obtain the generator polynomial of code from a sequence of noisy codewords. Bose–Chaudhuri–Hocquenghem (BCH) codes as a type of cyclic code, with large Hamming distance and error correction capability, are widely used in data storage and communication systems [2–5]. In Ref. [5], Wu et al. proved that their proposed method is advantageous for reconstruction of BCH codes in hard decision situation when the length code is known. To elucidate, the method proposed in Ref. [5] formulated and solved blind reconstruction problems based on consecutive roots of generator polynomial which is important mathematical properties of BCH codes and single-error correction of noisy codewords. Using the innovative approach, erroneous codewords are eliminated which makes the method superior to other methods. However, the proposed method suffers from lack of accuracy in the cases when the numbers of received codewords with single error are low. In fact, the single-error occurring depends on cross-over probability of binary symmetric channel (BSC). As known, the probability of the single-error occurring will be decreased when the level of noise is critical. In addition, increasing the number of received codewords (while keeping the cross-over probability of the BSC the same) offers an advantage in blind reconstruction and multiple error correction of BCH codes. However, it has no further effect on improving performance once the number of noiseless codewords or the error-correcting capability increases. Moreover, considering the results in Ref. [5], it is evident that the performance of the proposed method critically depends on the probability of having noiseless codewords and single-error codewords. In fact, the performance of the method drastically decreases when the numbers of codewords that are multiples of the generator polynomial—noiseless codewords—and single-error codewords are below a certain number (which varies from case to case), regardless of how many random codewords may be multiples of the generator polynomial. Note that the difference between the aforementioned certain number and the random codewords is noticeable, and a considerable amount of data is missed using this method.

In this letter, we propose a low-complexity method for blind reconstruction of BCH codes that addresses these limitations. By introducing a new set of polynomials as potential generators for BCH codes, we evaluate this set under random conditions to identify the target generator polynomial. Monte Carlo simulations demonstrate the superiority of the proposed method compared to the approach in Ref. [5].

**System model and assumptions:** To comprehensively understand the scheme of our methodology, first, system model needs to be defined. Consider a binary BCH code with length  $n$  and dimension  $k$  ( $k < n$ )

where error-correcting capability is  $t$  (i.e. corrects up to  $t$  errors). The generator polynomial of  $BCH(n, k)$  can be written as [5]

$$g(x) = LCM[M_{\alpha^b}(x), M_{\alpha^{b+1}}(x), \dots, M_{\alpha^{b+2t-1}}(x)] \quad (1)$$

where  $LCM$  is least common multiple function and  $b$  is an integer ( $0 < b < n$ ).  $M_{\alpha^i}$  where  $0 \leq i \leq 2^m - 1$  is the minimal polynomial of  $\alpha^i$  over  $GF(2)$  [5].

The code length is assumed to be known, and the perfect codeword synchronization is achieved at the receiver. Let us assume that  $M$  noisy codewords are received. We denoted the  $j$ th received codeword by  $r_j$ ;  $j = 1, 2, \dots, M$ . The codewords can be expressed, respectively, in a vector or polynomial represented as follows:

$$r_j = (r_{j,0}, r_{j,1}, \dots, r_{j,n-1}) \quad (2)$$

$$r_j(x) = r_{j,0} + r_{j,1}x + \dots + r_{j,n-1}x^{n-1} \quad (3)$$

Note that if the  $r_j(x)$  is noiseless, the received codeword  $r_j(x)$  is divisible by  $g(x)$ . As a preliminary result, by finding consecutive roots of all received codewords, first step of reconstructing the generator polynomial is prepared.

The Galois field Fourier transform ( $GFFT$ ) is defined in Ref. [5] as follows:

$$r(x) = r_0 + r_1x + \dots + r_{n-1}x^{n-1} \quad (4)$$

$$R(X) = R_0 + R_1X + \dots + R_{n-1}X^{n-1} \quad (5)$$

$$R_j = r(\alpha^j) = \sum_{i=0}^{n-1} r_i \alpha^{ij} \quad (6)$$

where  $v(x)$  is a polynomial over  $GF(2)$ ,  $V(X)$  is  $GFFT$  of  $v(x)$ , and coefficients of  $V(X)$  is  $V_j$  ( $j = 0, 1, \dots, n-1$ ). One property of  $GFFT$  is that if  $\alpha^j$  is a root of  $v(x)$ , then the  $j$ th coefficient of  $V(X)$  is zero. Consequently, by applying  $GFFT$  on all received codewords, we can find  $V_j = 0$  where  $j = 0, 1, \dots, n-1$  and finally find roots of  $r_j(x)$ . Moreover,  $GFFT$  is performed as [5]

$$\bar{\mathbf{F}} = \begin{bmatrix} (\alpha^0)^0 & (\alpha^1)^0 & \dots & (\alpha^{n-1})^0 \\ (\alpha^0)^1 & (\alpha^1)^1 & \dots & (\alpha^{n-1})^1 \\ \vdots & \vdots & \ddots & \vdots \\ (\alpha^0)^{n-1} & (\alpha^1)^{n-1} & \dots & (\alpha^{n-1})^{n-1} \end{bmatrix} \quad (7)$$

$$\bar{\mathbf{R}}_{M \times n} = \begin{bmatrix} \bar{R}_1 \\ \vdots \\ \bar{R}_M \end{bmatrix} = \begin{bmatrix} \bar{r}_1 \\ \vdots \\ \bar{r}_M \end{bmatrix} \times \bar{\mathbf{F}} \quad (8)$$

where  $\bar{\mathbf{F}}$  is the discrete Fourier transform matrix,  $\bar{\mathbf{r}}_{M \times n}$  is received codewords matrix, and  $\bar{\mathbf{R}}_{M \times n}$  is  $GFFT$  of them.

Having in mind above information, by employing  $\bar{\mathbf{R}}_{M \times n}$  and consecutive roots of  $BCH$  generator polynomial properties, we introduce a new set of candidate generator polynomial of  $BCH$  codes. So that, each polynomial of the new set can be generator polynomial of  $BCH$  code when the code length is known and perfect synchronization is presented, similar to the assumption in Ref. [5].

**Proposed reconstruction method:** A reconstruction method of  $BCH$  codes for improving system performance is proposed in this section, which inherently eliminates noisy codewords and investigates all terms of the new set on random situation. At first, we discuss how to obtain the required new set.

**Set of candidate generator polynomials:** Obtaining the new set requires identifying the positions of consecutive roots in each received codeword. To achieve this, we define a new parameter that represents the position of the maximum length of consecutive roots as

$$A_j = \{b_j, b_j + 1, \dots, b_j + l_j - 1\} \quad (9)$$

where  $A_j$  shows the consecutive roots position of  $j$ -th row of  $\mathbf{R}_{M \times n}$ ,  $b_j (b > 0)$  is starting value of consecutive roots, and  $l_j (l > 1)$  is maximum length of consecutive roots [5]. Using above definition and Equation (1), the candidate generator polynomial is written as

$$f_j = LCM[M_{\alpha^{b_j}}(x), M_{\alpha^{b_j+1}}(x), \dots, M_{\alpha^{b_j+l_j-1}}(x)] \quad (10)$$

where  $f_j$  is candidate generator polynomial of  $j$ th received codewords, which can be zero or duplicate for  $j = 1, 2, \dots, M$ . Consequently, a new set is formed from prime candidate generator polynomials  $f_j$  ( $j = 1, 2, \dots, M$ ) as follows:

$$S = \{g_i(x); i = 1, 2, \dots, N\} \quad (11)$$

where each polynomial has distinct iteration. Therefore, for  $i = 1, 2, \dots, N$ , the iteration value  $\alpha_i$  is defined as

$$\alpha_i = \sum_{j=1}^M T(g_i(x), f_j(x)) \quad (12)$$

where  $T(g_i(x), f_j(x))$  is defined as

$$T_{i,j}(g_i(x), f_j(x)) = \begin{cases} 1 & \text{if } g_i(x) = f_j(x) \\ 0 & \text{if } g_i(x) \neq f_j(x) \end{cases} \quad (13)$$

*Distribution of received bitstream for candidate polynomial of new set:* Now, we provide a method to determine which of these candidate polynomials is the desirable generator polynomial of the received bitstream. Unlike previous methods that rely on the probability that a candidate polynomial could be the generator polynomial for the codewords, the proposed method considers the probability that the candidate polynomial is incorrect.

To do this, we introduce the concept of the ‘random situation’: It is considered the scenario when the chosen generator polynomial is incorrect, causing the received bitstream to behave like a random sequence, even in the case of noiseless scenario. This happens because the incorrect polynomial fails to reproduce the original codewords, leading to unpredictable bitstream behaviour. Based on whether  $r_j(x)$  is divisible by  $f_j(x)$ ,  $T_{i,j}$  in Equation (13) is defined as a random variable as follows:

$$\hat{T}_{i,j} = \begin{cases} 1 & \text{if } r_j(x) \text{ dividable to } g_i(x) \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

Consequently, the iteration value  $\alpha_i$  in Equation (12) based on the  $\hat{T}_{i,j}$  will be as iteration variable  $\hat{\alpha}_i$ :

$$\hat{\alpha}_i = \sum_{j=1}^M T(g_i(x), r_j(x)) \quad (15)$$

In random situation, probability of the  $r_j(x)$  being dividable to  $g_i(x)$  is  $P_{i,j} = \frac{1}{2^{\deg(g_i(x))}}$  [3]. Therefore, the distribution of  $\hat{T}_{i,j}$  is as follows:

$$\hat{T}_{i,j} \sim \text{Bernoulli} \left( P_{i,j} = \frac{1}{2^{\deg(g_i(x))}} \right) \quad (16)$$

where  $1 \leq i \leq N$ . Needless to say, that all these random variables  $\hat{T}_{i,j}$  are independent from each other and identically distributed (i.i.d.). It is known that the  $P_{i,j}$  is smaller than the probability of the  $r_j(x)$  divided by  $g_i(x)$  when the  $g_i(x)$  is the generator polynomial of received codewords. Afterwards, iteration variable  $\hat{\alpha}_i$  will be a random variable with Gaussian distribution  $N(M \times P_r, M \times P_r(1 - P_r))$  by Central Limit Theorem [3].

Then, we introduce a threshold  $\tau_i$  for decision-making to determine whether  $g_i(x)$  is not in a random situation and can be considered a valid generator polynomial. This is based on the distribution of  $\hat{\alpha}_i$  as follows:

Let  $P_{fa}$  represent the probability of a false decision, that is the minimum probability that the candidate polynomial and received bitstream are in a random situation. As the distribution of  $\hat{\alpha}_i$  under random situation is known, we choose threshold  $\tau$  such that  $P_{fa} = P_r \llbracket \hat{\alpha}_i \leq \tau_i \rrbracket = \alpha$

### Algorithm 1 Blind reconstruction of BCH codes

**Inputs:** Mnoisy binary codewords  $r_j$ ;  $j = 1, 2, \dots, M$  with length  $n$ , where none of them is all-zeros.

**Step 1:** For  $j = 1, 2, \dots, M$

• Find GFFT of all received codewords using Equation (8).

**Step 2:** Form and arrange the set of prime candidate generator polynomials in sequentially upward order as  $\{g_i(x); i = 1, 2, \dots, N\}$ .

**Step 3:** For  $i = 1, 2, \dots, N$

• Calculate iteration values  $\alpha_i$  using Equation (12).

• Make Decision using Equation (17).

• If  $\alpha_i$  is not in random situation, End For.

**Output:** Return the generator polynomial  $\hat{g}(x) = g_i(x)$ .

**Table 1.** Candidate generator polynomials and their iteration order in one simulation round for BCH(31, 16) and  $g(x) = x^{15} + x^{11} + x^{10} + x^9 + x^8 + x^7 + x^5 + x^3 + x^2 + x + 1$

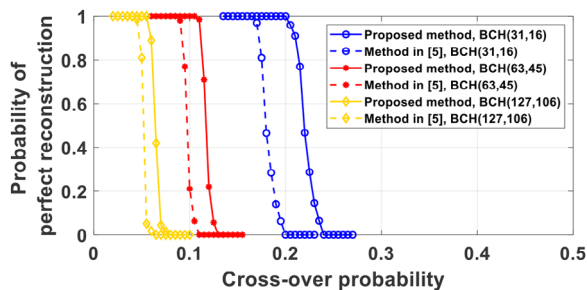
| $i$ | $g_i(x)$                                                                           | $\alpha_i$ | $\lceil \tau_i \rceil + 1$ |
|-----|------------------------------------------------------------------------------------|------------|----------------------------|
| 1   | $g_1(x) = 1 + x^2 + x^5$                                                           | 310        | 362                        |
| 2   | $g_2(x) = 1 + x^3 + x^5$                                                           | 312        | 362                        |
| 3   | $g_3(x) = 1 + x + x^2 + x^4 + x^5$                                                 | 311        | 362                        |
| 4   | $g_4(x) = 1 + x + x^3 + x^4 + x^5$                                                 | 310        | 362                        |
| 5   | $g_5(x) = 1 + x^4 + x^5 + x^6 + x^{10}$                                            | 11         | 15                         |
| 6   | $g_6(x) = 1 + x^3 + x^4 + x^9 + x^{10}$                                            | 9          | 15                         |
| 7   | $g_7(x) = 1 + x^2 + x^3 + x^5 + x^7 + x^8 + x^{10}$                                | 10         | 15                         |
| 8   | $g_8(x) = 1 + x^3 + x^5 + x^6 + x^8 + x^9 + x^{10}$                                | 9          | 15                         |
| 9   | $g_9(x) = 1 + x^2 + x^3 + x^4 + x^6 + x^7 + x^8 + x^9 + x^{10}$                    | 9          | 15                         |
| 10  | $g_{10}(x) = 1 + x + x^2 + x^3 + x^5 + x^7 + x^8 + x^9 + x^{10} + x^{11} + x^{15}$ | 22         | 3                          |
| 11  | $g_{11}(x) = 1 + x + x^6 + x^8 + x^{11} + x^{13} + x^{15}$                         | 1          | 3                          |

where  $\alpha \in (0, 1)$ . Details regarding the selection of this threshold can be found in Ref. [6]. The decision-making is as follows:

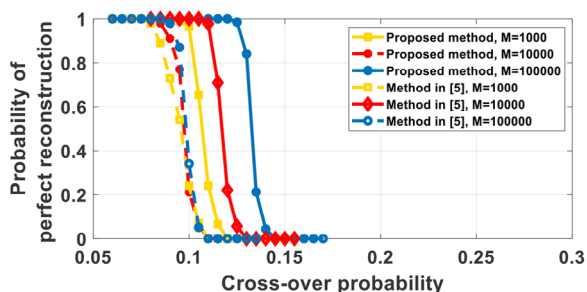
$$\text{Decision} = \begin{cases} \text{wrong } g_i(x) & \text{if } \alpha_i \leq \lceil \tau_i \rceil + 1 \\ \text{right } g_i(x) & \text{if } \alpha_i > \lceil \tau_i \rceil + 1 \end{cases} \quad (17)$$

Based on this observation, the proposed blind reconstruction algorithm is presented in Algorithm 1.

**Simulation results:** To validate our method, we simulated the algorithm's performance using Monte Carlo simulations. We compare it to the method in Ref. [5] using a binary BCH(31, 16) code with  $g(x) = x^{15} + x^{11} + x^{10} + x^9 + x^8 + x^7 + x^5 + x^3 + x^2 + x + 1$ . We generated 10,000 codewords transmitted through a BSC with a cross-over probability of  $p = 0.16$  and the false alarm probability of  $\alpha = 0.001$ . In our simulation, we received 22 noiseless and 144 single-error codewords. Using our proposed algorithm, we identified the candidate generator polynomial  $g_i(x)$ . As shown in Table 1, 11 prime candidate polynomials were obtained. The algorithm correctly identified the generator polynomial at  $i = 10$ , where  $\alpha_i$  exceeded the threshold  $\tau_i$ . On the other hand, the method from Ref. [5] incorrectly returns  $g_2(x)$  as the generator polynomial, relying on the most frequent consecutive roots among received codewords, which is prone to error when noiseless or single-error codewords are insufficient in this case  $p = 0.16$ . In simulations, for each value of cross-over probability  $p$ , Algorithm 1 is run 1000 times for a specific BCH code. The input of the algorithm is randomly generated in each simulation step. The number of correct reconstructions is then divided by 1000 to estimate the probability of perfect reconstruction while the probability of false alarm is set to  $\alpha = 0.001$ . Moreover, the method in Ref. [5] is also performed for each cross-over probability  $p$  using



**Fig. 1** Probability of perfect reconstruction for three different Bose–Chaudhuri–Hocquenghem (BCH) codes using the proposed method and the method in [5]



**Fig. 2** Performance of the proposed method for BCH(63, 39) code when the number of received code words are 1000, 10,000 and 100,000

1000 random simulation steps. In each simulation round,  $M = 10,000$  noisy codewords are generated and passed to the algorithms. To compare our proposed algorithm and the method in Ref. [5], three different BCH codes are considered:  $BCH(31, 16)$ ,  $BCH(63, 45)$  and  $BCH(127, 106)$ . Simulation results are shown in Figure 1. According to these results, our proposed method outperforms the method in [5], which relies heavily on the probability of receiving noiseless and single-error codewords. Moreover, performance of the proposed algorithm relies on the number of the noiseless received codewords. In fact, in a noisy channel, by increasing the number of received codewords in a specific value of the cross-over probability  $p$ , one can increase the probability of perfect reconstruction. This is confirmed by the simulation results of  $BCH(63, 39)$  presented in Figure 2. For example, in  $p = 0.11$ , when  $M = 1000, 10,000, 100,000$  codewords are received, the expected number of noiseless codewords is about 1, 7 and 64. As can be seen in the figure, the performance of the proposed method is improved from about 6%–100%. In contrast, the performance of method [5] is not significantly improved with increasing the number of received codewords.

**Conclusion:** In this letter, the problem of blind reconstruction of generator polynomial for BCH codes is investigated. Considering the fact that for an incorrect generator polynomial the received bitstream will be a random sequence for it, we obtained a new set of candidate generator polynomials using GFFT. Then, by investigating each of these candidates for random situation, the correct generator polynomial is found. Monte Carlo simulations along with analysis confirm the superiority of the proposed reconstruction algorithm compared to the latest presented method.

**Author contributions:** **Ali Sharifian Mazraeh Mollaei:** Conceptualization; investigation; methodology; writing the original draft. **Masoud Sharifian Mazraeh Mollaei:** Conceptualization; supervision; funding; review and editing.

**Conflict of interest statement:** The authors declare no conflicts of interest.

**Data availability statement:** The data that support the findings of this study are available from the corresponding author upon reasonable request.

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